

WatchDog AI

Project Team

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Chapter 1

Introduction

The WatchDog AI surveillance system is an automatic violent activity detection and criminal recognition system. The advanced system can also generate AI reports based on scene understanding and pin-point the exact location of the person who committed the violent act. This document is intended for developers, users, testers, and project managers although it can also be read and easily understood by a non-technical person. [1].

1.1 Problem Statement

Our traditional CCTV systems are able to capture and store the footage, however, they are unable to analyze it in real-time. Moreover, our current systems are manually monitored, resulting in human error and fatigue, leading to missed threats and delayed responses. Even our current AI systems have limitations, whether that is in accuracy, adaptability or real-time processing. To solve and automate all of these existing issues, we are developing an advanced model, capable of detecting violence and criminal activities, detecting and recognizing criminals using facial recognition, and implementing scene understanding to automatically generate reports using language models. This software will help in the timely detection of violent activities and criminals, improving security and providing a safer environment for everyone.

1.2 Scope

Due to the high importance and sensitivity of an advanced surveillance system, our software will have multiple functionalities in the scope.

1. The software will be detecting a broad range of criminal and violent activities, like weapons, abuse, arson, burglary, shooting, stealing, vandalism, violence.
2. It will also be trained to detect and recognize criminals (using facial recognition), who are known to be associated with criminal activities or who are on a watch-list.
3. The software will be capable of pin-pointing the exact location of the crime in a public/crowded place.
4. It will reduce the need to check the complete CCTV footage, by bookmarking the times of the violent activities.
5. Scene understanding will also be integrated in this software.
6. Automatic report generation will also be integrated in the software.
7. The system will also be providing real time and instant alerts to the user and authorities (based on the severity of the crime).
8. The software will provide the user with a web based app for monitoring the footage and surveillance.
9. The system will be integrated with existing security systems, allowing a seamless transition and operation.

1.3 Modules

The modules of the proposed project have been listed below. Each module highlights the features.

1.3.1 Module 1

1. Train multiple models to detect criminal and violent activities.
2. Implement facial recognition to recognize and detect criminals.

1.3.2 Module 2

1. Implement scene understanding and integrate automatic report generation.
2. Create a user friendly web app, to allow the user to monitor CCTV footage.

1.4 User Classes and Characteristics

Various user classes anticipated to be used in this product, and the description of their pertinent characteristics:

User class	Description
Violence	The class "Violence" will be used to train all the models that would be detecting weapons and violent activities in a CCTV footage.
Facial detection	The class "Facial detection" will be using models to detect and recognise criminals in the feed.
Scene understanding	The class "Scene understanding" will incorporate large vision models for scene understanding, which will be used by the "Report generation" class.
Report generation	The class "Report generation" would use the scene understanding results for automatic report generation.
Web app	The class "Web app" will allow the user to monitor the CCTV footage from multiple cameras and also alert the user if a violent activity is detected.

Table 1.1: Description of User Classes

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Event Response Table

Event	System Response
User login	The system will allow the user to enter their credentials and login to the web app of the application.
User clicks the "Next/Prev camera" button	The system shifts to the next or previous camera, respectively, and displays the CCTV footage from that camera. The user can click these 2 buttons infinite number of times and the display will start showing footage from the next/previous CCTV camera.
System detects a violent activity	The system will immediately alert the user about the activity detected and its type.
System detects a criminal	The system will immediately alert the user and display the details of the criminal on the web-page, allowing the user to alert the authorities (if needed).
User clicks on "Report generation"	The system will automatically generate a report for the user, containing details of the event occurred along with the scene understanding done by the model.
User clicks on "Book-marks"	The system will show the time periods that were book marked for detecting any sort of suspicious activity.

Table 2.1: Event Response Table

2.2 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style. This section is typically organized by feature as a system feature name and specific functional requirements associated with this feature. It is just one possible way to arrange them. Other organizational options include arranging functional requirements by use case, process flow, mode of operation, user class, stimulus, and response depend on what kind of technique has been used to understand functional requirements. Hierarchical combinations of these elements are also possible, such as use cases within user classes.

2.2.1 Module 1

Following are the requirements for module 1:

1. The system will be capable of detecting violent activities in a CCTV footage.
2. The system will immediately be able to detect any criminals in a CCTV footage.

2.2.2 Module 2

Following are the requirements for module 2:

1. The system will be able to do scene understanding and generate reports based on its understandings.
2. The backend will be integrated with the frontend and a web app will be delivered as a final product.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements.

2.3.1 Reliability

In case of a loss of signal/connectivity, the system may fail for a few seconds as that specific CCTV camera will be unavailable for recording any footage. This, however, is very rare to happen and may only occur in case of damage to the hardware of the CCTV camera or any cut-off in power lines. If the system is not backed up in case of power outages, it will not be available until the power is restored. The availability of a backup power source can resolve this issue. Moreover, the system will fail to detect any violent activities if the view is blocked by an object.

2.3.2 Usability

The system will be easy to use for not only a CCTV inspector, but anyone who wishes to check the footage's. The feed from camera(s) will be displayed on the screen and the user can go onto the next or previous cameras by clicking on their respective buttons. A button will also be available for automatic report generation from the trained model. The web app will be easy to navigate and control, and will not require any technical expertise.

2.3.3 Performance

No specific performance requirements needed for the system operations except for a web monitor. The system can be integrated with any CCTV camera, thus, there is no specific requirement. However, for faster detections and to avoid any delays, it is recommended to have a system with at least 16GB of RAM and a GPU.

2.3.4 Security

In order to prevent the system being hacked or any data leakage, it is essential that the user keeps their credentials confidential. Moreover, the criminal dataset (for criminal detection) should not be exposed and its database access should not be with every user as the privacy of these individuals is a major concern and threat.

2.4 Domain Model

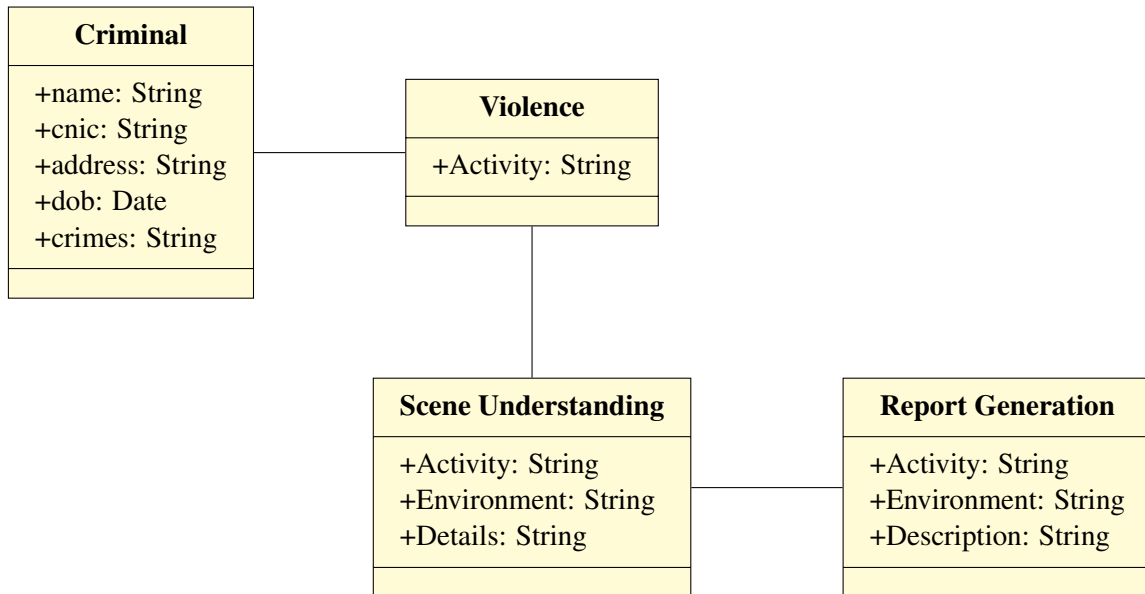


Figure 2.1: UML Diagram

Use-case diagram:

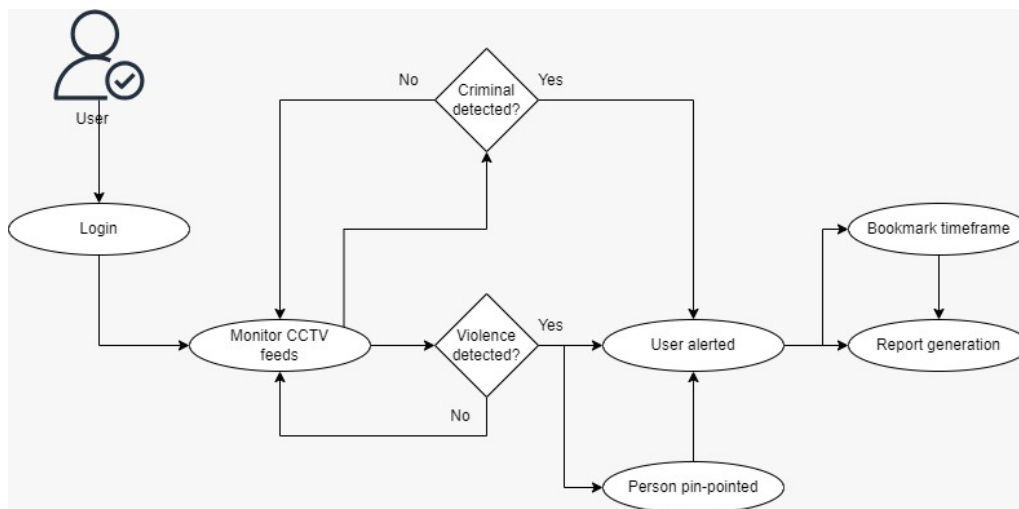


Figure 2.2: Use-case Diagram

Chapter 3

System Overview

3.1 Architectural Design

In the first module, we'll start by creating multi-models for violent activities detection. These models will automatically detect any activity that happens in the CCTV footage. This subsystem will be integrated with another subsystem, that will be detecting criminals that appear in the CCTV footage. In the second module, we'll be creating a subsystem for scene detection. This system will be integrated with the previous two, as it will be using the results from the trained models for scene understanding. The scene understanding subsystem will be merged with the subsystem for report generation, which would be using the results of scene understanding for automatically generating a report. Lastly, a subsystem for pin-pointing the exact person who was involved in the violent activity, will be blended with the subsystem for scene understanding.

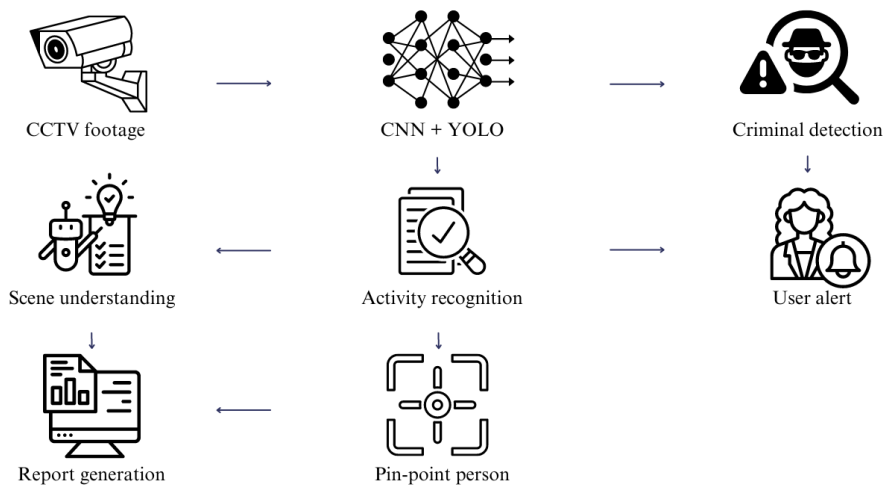


Figure 3.1: Architecture Diagram

3.2 Design Model

Design Model for Procedural Approach

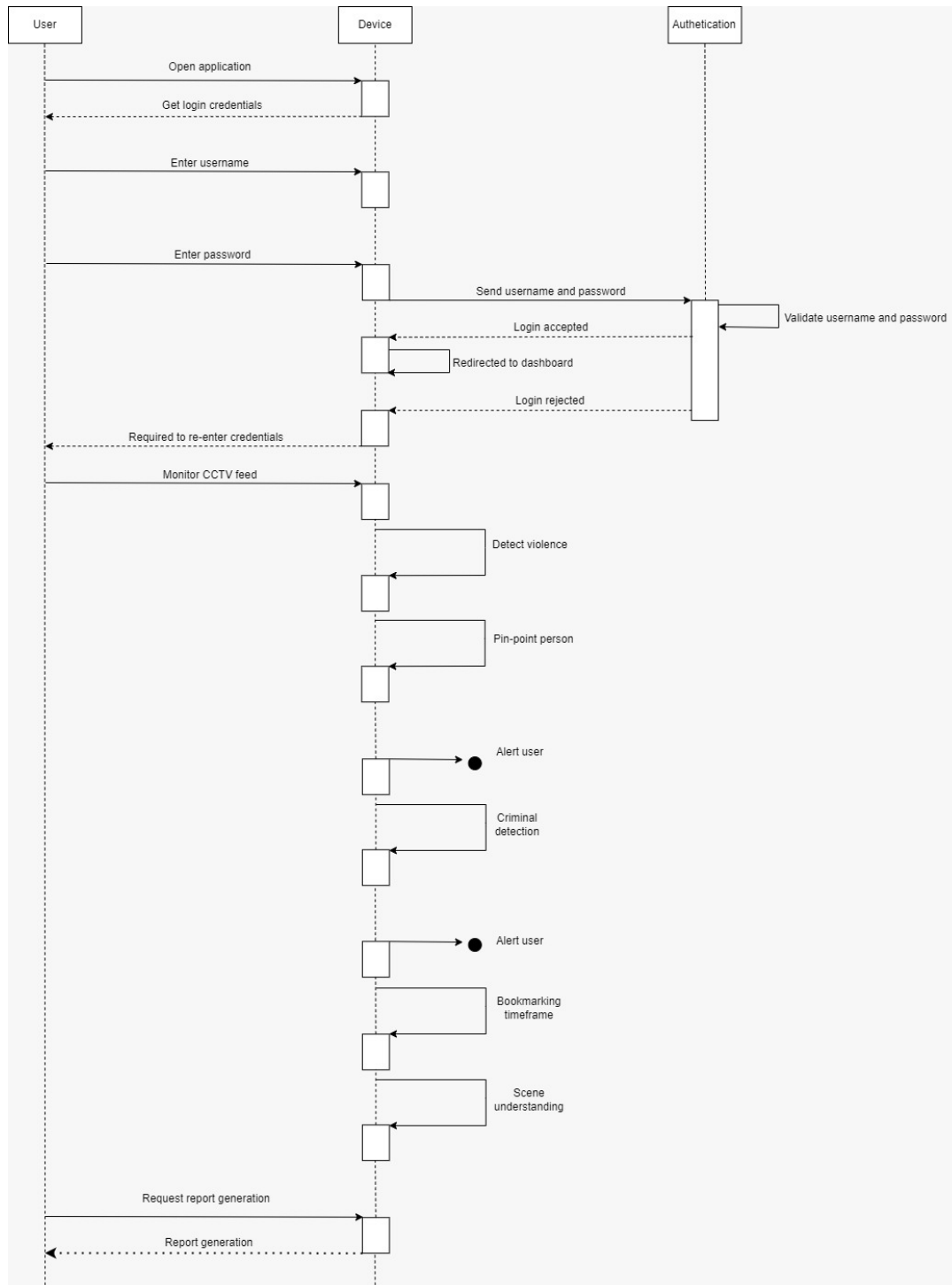


Figure 3.2: Sequence Diagram

3.3 Data Design

Criminal dataset, which consists of criminals and their details, will be stored in a database, and will be used for detecting criminals in a CCTV footage, in real time. This database will be integrated with the facial detection subsystem. Only a few members of the development team (not the end user) will have access to this database, ensuring that there is no compromise on security. Any other images that are used for model training will simply be downloaded on the local system and will be used to train and test the model on.

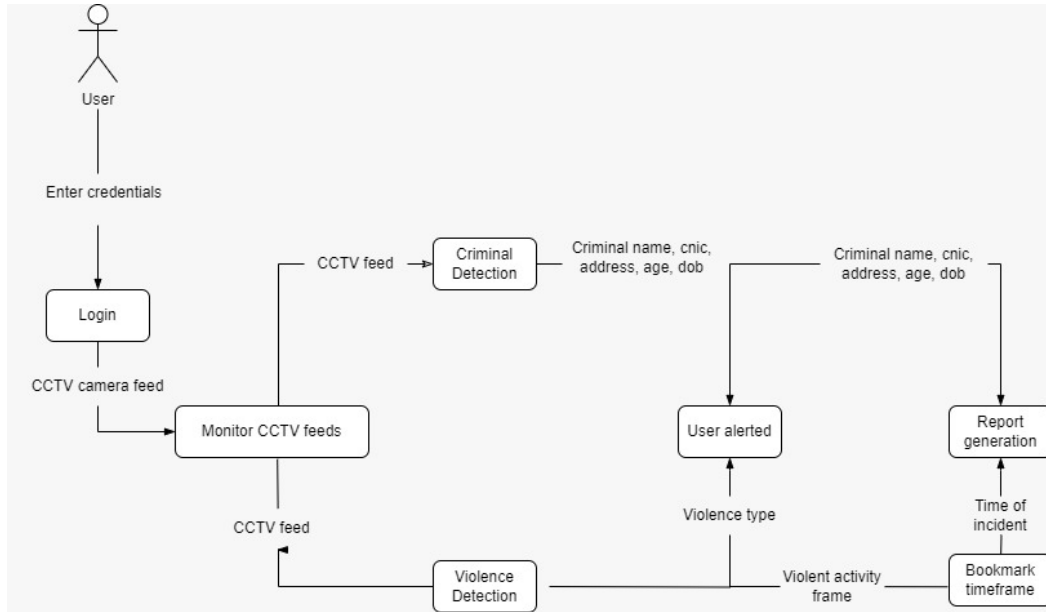


Figure 3.3: Dataflow Diagram

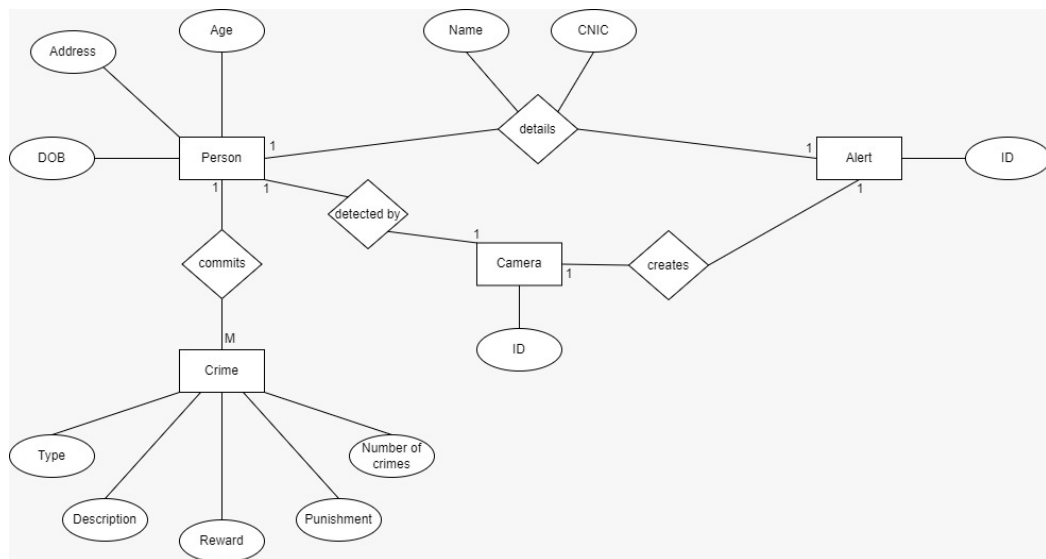


Figure 3.4: ERD Diagram

Bibliography

- [1] R Tavassoli L Harris S Tan J Waite, J Feng and S Chakraborty. Active shooter detection and robust tracking utilizing supplemental synthetic data. *Arxiv*, 2023.